

Loan Approval Prediction

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Executive Summary

Loan approval is the process by which financial institutions, such as banks or credit unions, evaluate and decide whether to grant a loan to an individual or entity. This decision is based on an assessment of the applicant's financial health, credit history, ability to repay the loan, and numerous other factors. In a loan approval prediction project, the goal is to build a machine learning model that predicts whether a loan application will be approved or rejected based on various features of the applicant and loan details. Financial institutions want to automate and optimize this decision-making process to save time, reduce costs, and minimize risks associated with loan defaults.

The dataset used in this project is obtained from the Kaggle website (<https://www.kaggle.com/datasets/itshappy/ps4e9-original-data-loan-approval-prediction>). It is a simulated dataset based on the information obtained from various financial institutions. This dataset contains 32581 loan applications with truth labels showing only 7108 approvals as opposed to 25473 rejections. 11 variables collected in the loan application process are used in model training. The imbalance between the approvals and rejections makes it challenging to predict approvals accurately as there are more rejections than approvals.

5 different supervising models were considered and numerous iterations were done to understand the most important features, underlying relations within the predictors and the impact of their interactions on the target variable. Studies are done to find the optimal hyperparameters. Considering the accuracy, XGBoost model performed the best. It has a sensitivity value of 0.993 and specificity of 0.73 and a kappa value of 0.795. However, it comes with its own set of limitations as it requires considerable time for hyperparameter tuning, which may limit its real-time application capabilities.

For the future, these predictive models can be improved using criteria such as kappa values and computation time. Kappa values are more useful when the dataset is imbalanced. Accuracy was considered for the overall model performance in this project, which works well for overall performance however it is not the best for the dataset presented. Thus, random forest model can be explored further. Also, it makes sense to focus using the most important features and develop new features taking into account the correlations between various features. The XGBoost model needs to be tested on a validation dataset to study the model's performance and improve where needed to be able to use in the future. Another suggestion for future would be to keep updating the model as and when more data becomes available in the future.

Introduction to the problem

Loan approval is a critical process for financial institutions, directly impacting their ability to manage risk and allocate financial resources efficiently. Evaluating loan applications involves assessing the credit risk associated with applicants based on several factors, including their financial standing, credit history, and demographic information. Making accurate loan approval decisions not only safeguards the financial stability of the lender but also ensures fair access to credit for eligible applicants.

The **Loan Approval Project** aims to analyze a comprehensive dataset containing detailed credit risk and loan application information. By developing a machine learning model, this project seeks to classify loan applications as "approved" or "not approved." The dataset includes metadata and critical attributes about applicants that influence loan approval decisions. Through exploratory data analysis (EDA) and advanced predictive modeling, this project will provide insights into key drivers of loan approval and optimize decision-making processes.

Data Description

The dataset has 11 predictors and 1 target variables. Among the 11 predictors, 7 are numeric variables and 4 are categorical variables.

Key Highlights of the Dataset:

1. Applicant Demographics

- Variables such as "age" provide context to the profile of loan applicants.

2. Financial Attributes

- Features like "annual income", "home ownership type", and "employment length" help gauge the applicant's financial health.

3. Credit History

- Data on "credit history length", "default history", and other repayment-related variables serve as indicators of an applicant's creditworthiness.

4. Loan Details

- Information such as "purpose of the loan", "loan amount", "loan interest rate" and "loan grade" add depth to the decision-making process.

5. Loan Status

- The target variable or truth label, indicates whether the loan application was approved (typically encoded as "1") or rejected (typically encoded as "0").

Exploratory Data Analysis

An exploratory data visualization was done for better understanding of the data, study outliers, skewness and get a sense of underlying relations within the predictors themselves and of the predictors with the target variable. Several visualizations were looked at and those that are more useful to understand the problem are presented here.

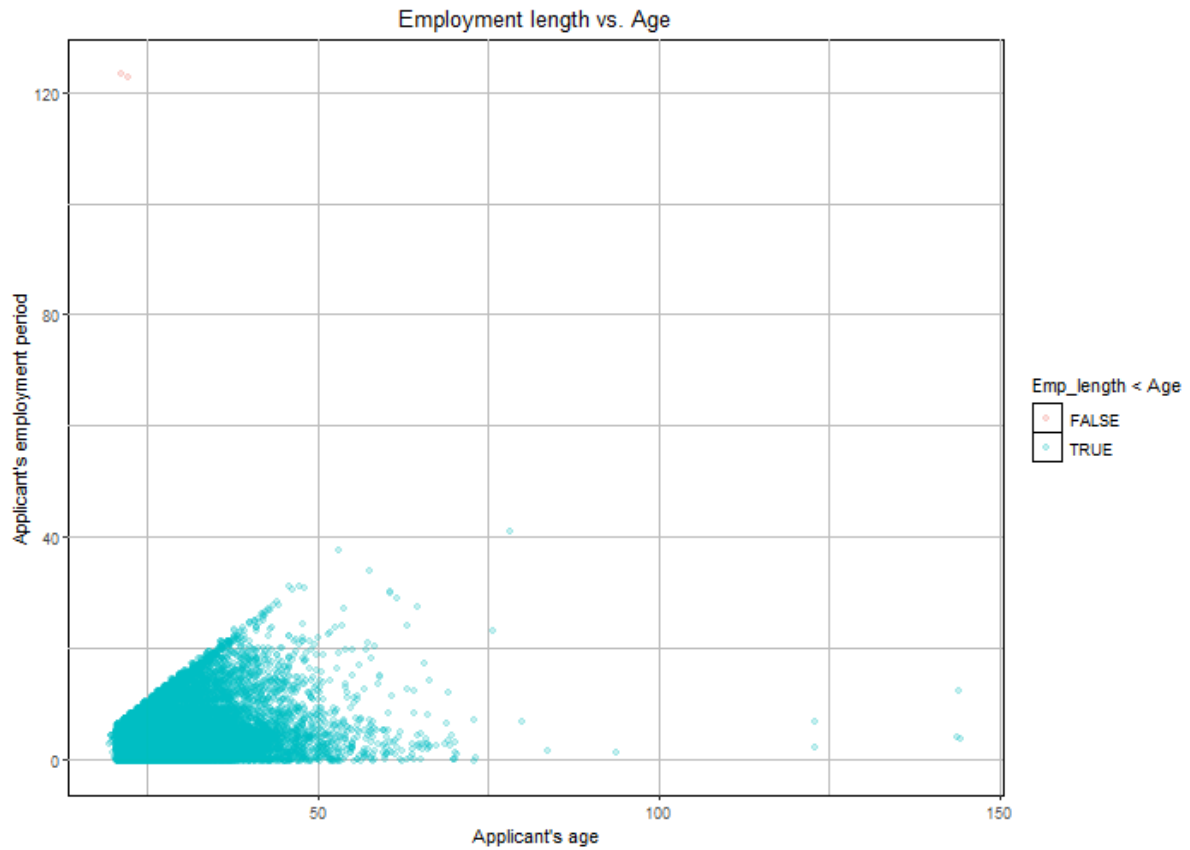


Figure 1: Finding outliers

This graph shows the relationship between applicant's employment period and age in years. Overall the distribution looks normal with some obvious extremities. One extremity is where the applicant's age is significantly higher and with low employment period. Right away, this doesn't call for any actions as these scenarios are possible in the real world, but it is the other extremity that one should be worried about. The second extremity is where a person's age is below 25 years, but their employment length is above 120 years. Now, this is an impossible situation to occur in reality and therefore, it makes sense to remove any and all instances where an applicant's age is smaller than their employment length.

It is also useful to note that there are some unrealistic numbers about a person's age, i.e. greater than 125. The oldest living person is known to be of 116 year and thus, datapoints corresponding to $\text{age} > 120$ can be removed.

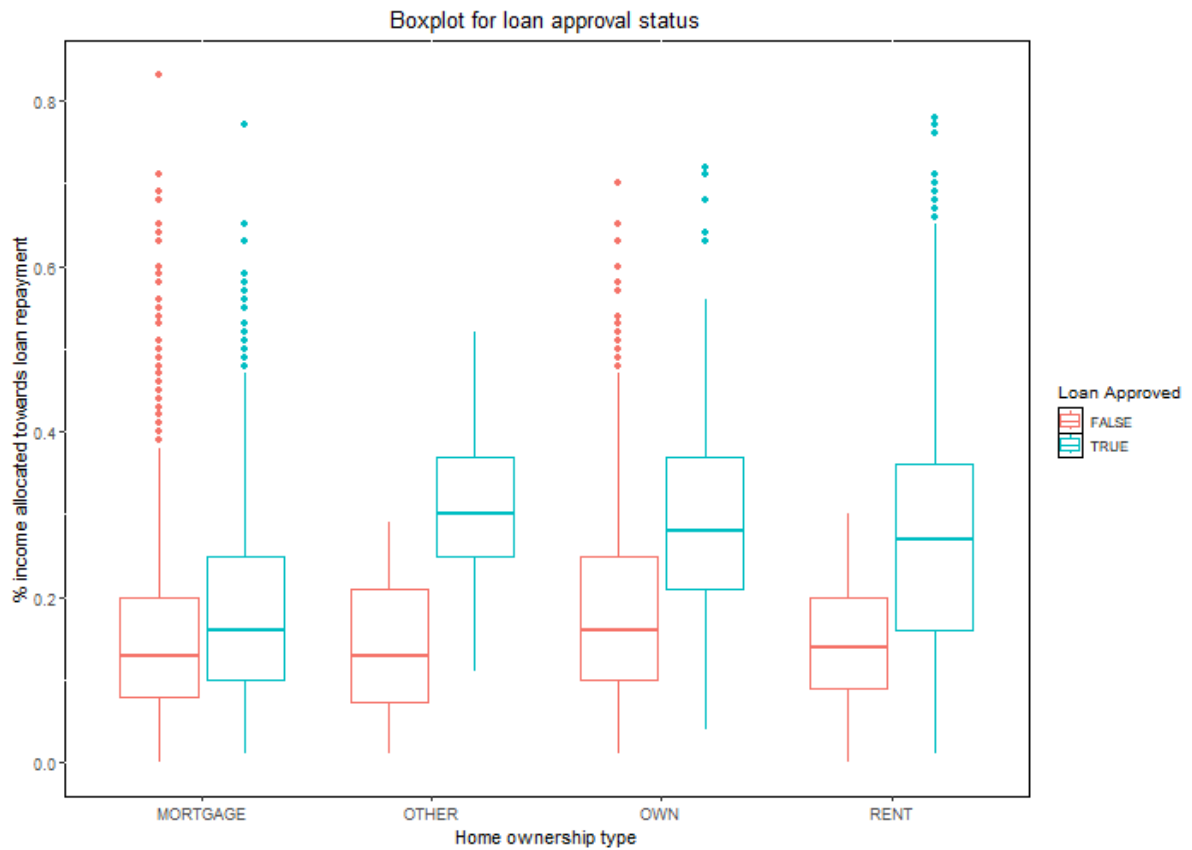


Figure 2: Boxplot for loan approval status

The above graph explores the relationship between the percentage of income allowed towards loan repayment and home ownership type of the applicant. These are two important variables that impact the target variable and the same is seen in the graph above. There is a clear relationship between the percentage of income allocated and loan approval status. In general, higher the allocated amount for each home ownership type, higher the chances for loan approval. There is also a direct relationship between the percent of income allocated versus home ownership type. In general, applicant's that either own or rent the home or those who fall under the **OTHER** category which could be living out a motel/van/airbnb, etc. are more likely to allocate more money towards loan repayment and hence have higher chances of getting them approved as well.

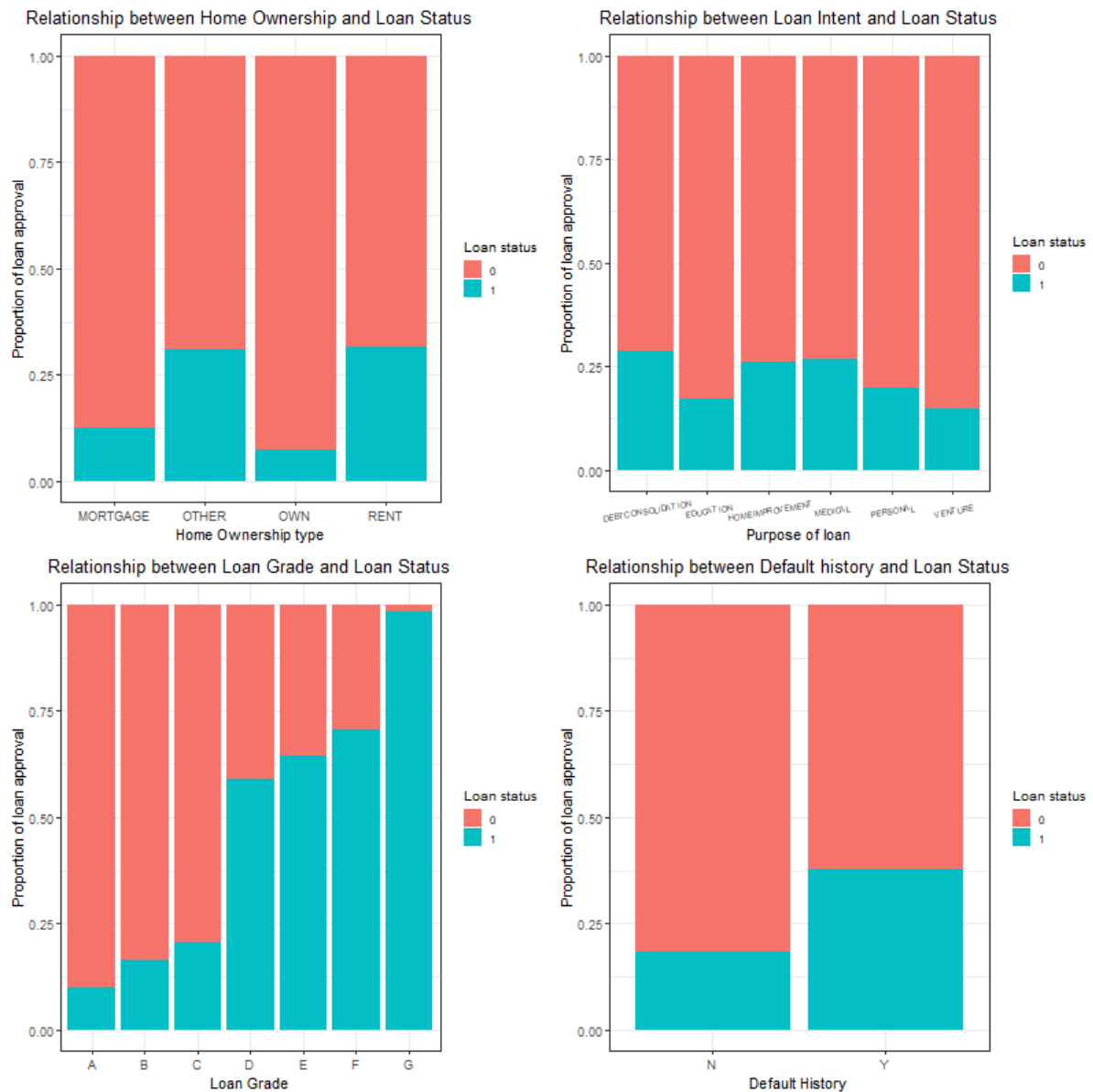


Figure 3: Relationship of loan approval with (Top left) homeownership type (Top right) loan intent (Bottom left) loan grade (Bottom right) default history

The above set of plots in understanding the relationship between the target variables and predictors.

- For applicants that either rent the home or fall under **Other** category, the loan approval rate is much higher compared to other two categories. Those who own home have lower approval rates, which is counterintuitive and hard to predict without doing the thorough analysis.
- Loan applications for debt consolidation, medical reasons and home improvements have higher approval rates compared to the personal loans, education loans and venture loans.
- Lower the loan grade, lower is the risk, however there is a negative correlation between the risk and loan approval. Higher the grade, higher the risk and also higher interest rate (from heatmap for correlation) and higher the chances for loan approval.
- If a person has a default history, their chances of loan approval are higher which is again counterintuitive. Hence, this visualization is useful to study that default history has direct impact on loan approval.

Data preprocessing

Missing Values

Only two variables `person_emp_length` (2.75%) and `loan_int_rate` (9.56%) have missing values. For **supervised learning**, the missing values were imputed using the predictive mean matching (PMM). For **unsupervised learning**, the missing values were imputed using the `kNN` technique by looking at 5 nearest neighbors using the Euclidian distance from the remaining non-missing values for those variables.

Outliers

- Data points where an applicant's employment length(in years) is greater than their actual age were removed.
- Data points corresponding to applicants over 120 years old were removed. The record for oldest living person till today is only 116 years, so it is reasonable to put a upper bound on applicant's age.

Feature transformation

Comprehensive Data Preparation Plan

1. Merge low-frequency categories

- Combine `person_home_ownership` values 'OWN' and 'OTHER' into a single category 'OTHER' due to low frequencies.
- Combine `loan_grade` values 'F' and 'G' into a single category 'OTHER' for the same reason.

2. Convert categorical variables to numerical format

- Convert `person_home_ownership` and `loan_intent` into numerical format using one-hot encoding or label encoding as appropriate.
- Convert `loan_grade` into an ordinal factor reflecting the grade hierarchy (e.g., 'A' < 'B' < 'C' < 'D' < 'E' < 'OTHER').
- Convert `cb_person_default_on_file` into an ordinal factor ('No' = 0, 'Yes' = 1).
- Convert the ordinal factor into a numerical variable.

3. Remove redundant columns

- Remove the original categorical columns: `person_home_ownership` , `loan_intent` , `loan_grade` , and `cb_person_default_on_file` , as their numerical representations have been created.

4. Apply logarithmic transformation to numerical variables
 - Apply logarithmic transformation to variables with skewed distributions to approximate normality: `person_age` , `person_income` , `person_emp_length` , `loan_amnt` , `loan_int_rate` , `cb_person_cred_hist_length` (rename as `log_cred_hist_length` and remove the original variable).
5. Add a new feature
 - Create `loan_percent_income` as the ratio of loan amount to applicant's annual income.
6. Standardize numerical variables
 - Standardize (z-score normalization) to have a mean of 0 and a standard deviation of 1 for use in distance-based models.
7. Remove zero-variance variables
 - Identify and remove any variables with zero variance, as they provide no information for modeling.
8. Filter variables based on correlation
 - Select variables with an absolute correlation coefficient greater than 0.2 to the target variable.
9. Explore interaction terms Create and evaluate interaction terms for potential improvements in predictive performance:
 - `loan_percent_income` * `log_income`
 - `log_int_rate` * `loan_grade_numeric`
 - `person_home_ownership_RENT` * `loan_percent_income`

Modelling Choice

For a loan approval prediction since it involves labeled data, it makes sense to use supervised learning model. Unsupervised learning model is useful when they are no labels and the goal is to find hidden patterns or anomalies.

An attempt was made to use unsupervised learning model. In particular, density based spatial clustering with noise (DBSCAN) was used to check if it predicts two clusters. The reason to choose DBSCAN clustering instead of k-means or hierarchical clustering is that there is no requirement to specify the number of clusters. Initially it appeared that the model predicted 2 clusters and several anomalies, however the values for evaluation metrics like precision, recall and F1 Score had perfect scores of 1 and raised some alarms. Further debugging led to the conclusion that truth labels were used accidentally in DBSCAN clustering and thus it resulted in two perfect clusters and some noise points/anomalies. These results are shown in the appendix and report will focus mainly on supervised learning models and their results.

Various supervised learning models were explored, for e.g. generalized linear model(GLM), Least Absolute Shrinkage and Selection Operator(LASSO), random forest, gradient boosting machine(GBM) and extreme gradient boosting(XGBoost). XGBoost was found to be the best performing model.

Model Validation Methods

To ensure the stability and predictive power of the model, 10-fold cross-validation and hyperparameter tuning were employed during modeling to reduce bias and improve generalization performance.

Instead of splitting the dataset into an arbitrary ratio for training and testing, a cross-validation method is applied as it ensures that the entire dataset is seen by the model and no bias is introduced when focussing only on a fixed sample of the dataset.

- 10-Fold Cross-Validation: Maximizes data utilization, reduces performance fluctuations caused by the division of training and testing datasets, improves the stability of model evaluation across different data splits, and minimizes model bias to enhance generalization performance.

Hyperparameter tuning

A random search hyperparameter tuning was used as it prevents data leakage during the tuning process, ensures the objectivity of performance evaluation, optimizes model hyperparameters, and evaluates model performance. This guarantees that the performance metrics after hyperparameter tuning are highly reliable.

Results for Supervised Learning Models

To comprehensively and objectively evaluate the performance of each model, the following metrics were employed: ROC, Accuracy, Kappa, Sensitivity, and Specificity. These metrics are not only applicable to classification problems but also assist in selecting the optimal model and explaining its suitability.

Phase 1 Modeling: GLM, LASSO, RANDOM FOREST, GBM

This was an initial set of modelling study done to check which models perform better and what variables or interactions are useful. This study was then used to decide which models need further improvement in the next phase.

Method	Setting	ROC	Sens	Spec	Accuracy	Kappa
GLM	All Variables + Interaction	0.888061	0.952276	0.585979	0.868389	0.590788
LASSO	All Variables + Interaction	0.888305	0.953115	0.579447	0.867543	0.586638
RANDOM FOREST	All Variables	0.940561	0.995763	0.735622	0.936182	0.801773
GBM	All Variables + Interaction	0.930197	0.989773	0.733581	0.931102	0.787502

Table 1: Modelling results for all models

Based on the data comparison above, GBM and RANDOM FOREST significantly outperformed GLM and LASSO in terms of ROC, Sensitivity (Sens), Specificity (Spec), Accuracy, and Kappa. Among them, RANDOM FOREST slightly outperformed GBM. In the next phase of improving the model's performance, only RANDOM FOREST, GBM and XGBoost were considered. The result of the same are shown in **table 2**

Phase 2 Modeling: GBM, RANDOM FOREST, XGBoost

Here only three models are considered and further improvements are made using hyperparameter tuning compared to phase 1 studies.

Method	Setting	ROC	Sens	Spec	Accuracy	Kappa
RANDOM FOREST	All Variables	0.940566	0.995763	0.735629	0.936181	0.801776
GBM	Random Search Hyperparameter Tuning	0.945343	0.990733	0.737162	0.935409	0.793518
XGBoost	Hyperparameter Tuning	0.950954	0.992775	0.733785	0.936268	0.795393

Table 2: Modelling results for GBM, random forest and XGBoost

The visual comparison chart is as follows :

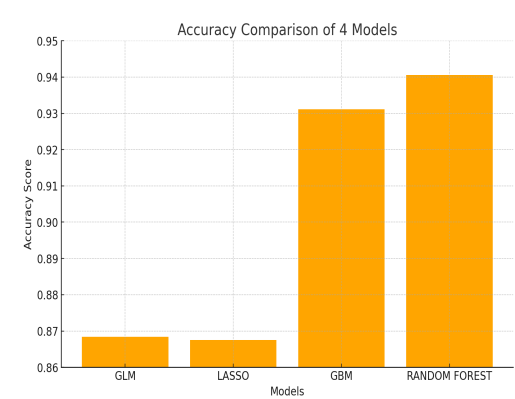


Figure 4: Accuracy value comparisons

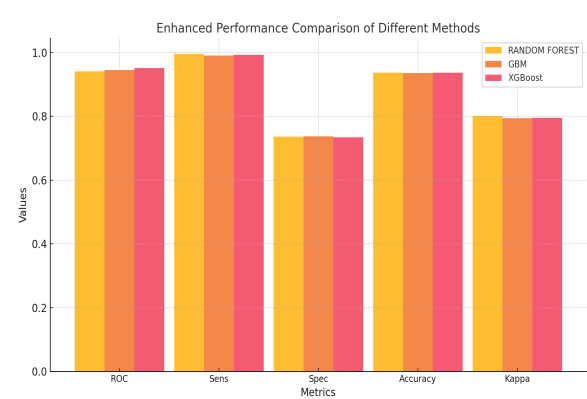


Figure 5: Performance comparisons

- XGBoost has the highest ROC value (0.950), indicating that it slightly outperforms other methods in terms of discrimination ability.
- RANDOM FOREST has the highest sensitivity (0.996), making it the best at detecting positive classes.
- GBM is slightly better in specificity (0.737), demonstrating a slight edge in correctly identifying negative classes.
- XGBoost and RANDOM FOREST have the highest accuracy, which is nearly identical (0.936), showcasing excellent overall classification performance.
- RANDOM FOREST has the highest Kappa value (0.802), indicating the best agreement between predictions and actual results.

Overall, **XGBoost performs best**, especially in ROC and Accuracy, while RANDOM FOREST excels in Sensitivity and Kappa. GBM also demonstrates solid overall performance but is slightly weaker than the others in some specific metrics

Results for XGBoost

Table 2 shows XGBoost's performance is very close to the optimal values across all metrics, making it the most well-rounded and balanced model.

The best model was obtained by using 10-fold cross validation technique. However, since we do not have a separate validation dataset, the entire dataset is used to check the performance of this model. The performance is studied with the help of ROC curve, confusion matrix, and residual analysis. Important features are also obtained using this model.

ROC Curve: The ROC curve demonstrates the performance of the classification model at different thresholds. The curve being close to the upper-left corner indicates excellent classification performance. The AUC = 0.993 (highlighted in red), shows that the model can almost perfectly distinguish between positive and negative samples. The curve's proximity to the upper-left corner also signifies a very low error rate in distinguishing between positive and negative samples. This ROC curve indicates that the XGBoost model has outstanding performance, with an AUC value close to 1 reflecting exceptionally strong classification ability. Comment - Since all data has been "seen" by the model during training, the performance appears overly optimistic..

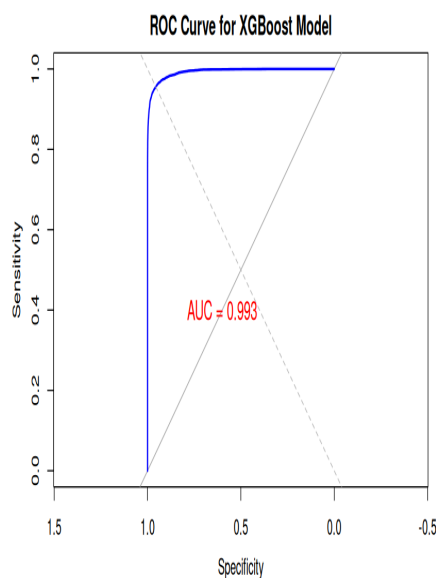


Figure 6: ROC curve

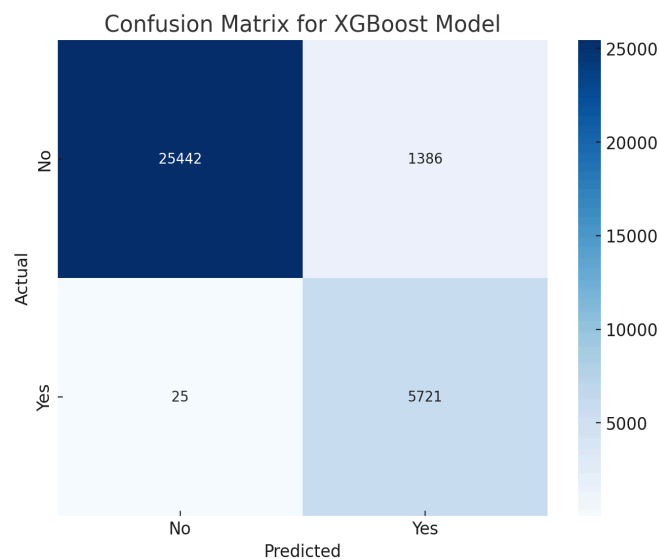


Figure 7: Confusion Matrix

Confusion Matrix

- Accuracy: 95.67%, showing a very high overall prediction accuracy.
- Sensitivity: 99.90%, indicating that the model predicts the "No" class with near-perfect accuracy and almost no missed cases.
- Specificity: 80.50%, suggesting that the model performs well in predicting the "Yes" class, but with a slightly higher false positive rate (1386 "No" samples misclassified as "Yes").
- Kappa Coefficient: 0.8636, indicating a very high level of agreement between the predicted and actual classifications, close to 1, which reflects good model performance.

Comment - With significantly uneven distribution of positive and negative samples, the confusion matrix tends to favor the majority class, resulting in an apparent improvement in overall performance.

Feature Importance

- The core variables `loan_percent_income` and `log_income` are the most important features, highlighting the relationship between borrowing and income as the primary basis for the model's decision-making.
- Other important features include `loan_grade_numeric` (loan rating), `cb_person_default_numeric` (default history), and `person_home_ownership` (homeownership status).

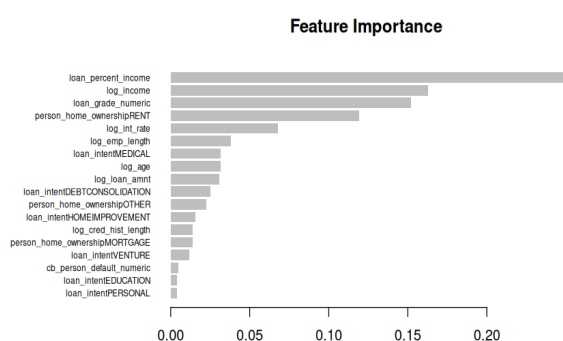


Figure 8: Ranking features by their importance

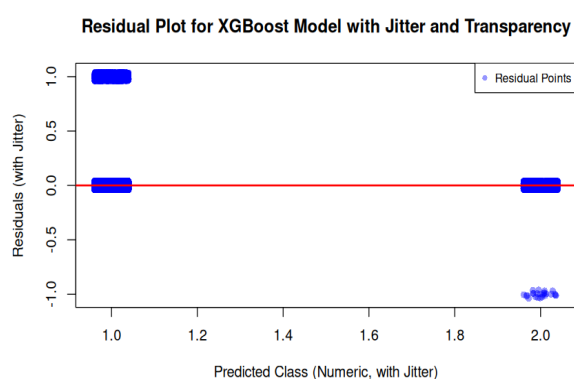


Figure 9: Residual Analysis

Residual Analysis

- Most data points have a residual of 0, indicating that the model predicts the majority of samples accurately.
- Misclassified residuals are within ± 1 , and the quantity is relatively small, suggesting that the overall model error is low.

Results for Unsupervised Learning

DBSCAN clustering was attempted by accidentally using the truth labels. Although, truth labels are not to be used for unsupervised learning, but the inferences from the clustering can still aid to make decisions in cases where there are anomalies or a need for further investigation for an application. The clustering results are discussed below and an attempt has been made to find patterns among these clusters as well as noise points.

DBSCAN clustering for 32579 objects.

Parameters: `eps = 2.3`, `minPts = 9`

Using euclidean distances and `borderpoints = TRUE`

The clustering contains 2 cluster(s) and 71 noise points.

```
0      1      2
71 25424 7084
```

These predictions are further compared with the existing target variable `loan_status` using the table below. The row value 0 corresponds to points where the loan was not approved and 1 corresponds to those applications where the loan was approved. DBSCAN clustering identifies these clusters such that it predicts the loan approval correctly ~99.68% of the times and it predicts the loan rejection correctly ~99.81% of the times.

0	1	2	
0	48	25424	0
1	23	0	7084

Inferences from clusters and noise points

In **cluster 1**, most applicants are young and in their 20's, the income is either very high or very low, the high income also corresponds to zero employment length in most cases implying that the person must own their own business, however, the most important feature is percentage of money allocated towards loan repayment which is low considering the high income in most cases.

In **cluster 2**, applicants are mostly young and in their 20's, with an moderate amount of income in most cases, relatively higher length of employment on average compared to cluster 1. However, the percentage of money allocated towards loan repayment is quite high compared to cluster 1.

For **noise points**, the applicants are either very old or of moderate age, in their 30's, with exceptionally high incomes and very low percent of money allocated towards loan repayment. Thus, these applications need further investigations since these are anomalies.

Truth labels were used to evaluate the model. The Adjusted Rand Index (ARI) was found to be 0.994 when using noise points meaning there is near perfect agreement between the clusters and truth labels. Precision=1, Recall=1 and F1 Score=1 was obtained by removing noise points meaning there are no false positives or negatives and the model accurately predicts the clusters. This appears too good to be true and a further investigation was done only to realize that the truth labels were accidentally used during the training. Thus, further work needs to be done to fine tune this modelling.

Conclusion and Future work

Summary of the problem

The goal of this project is to study a dataset containing loan applications and truth labels corresponding to approval/rejection and find a model to predict whether a new loan application be approved or rejected. This is a binary classification problem that can be solved using a supervised learning model as truth labels are present. However, special care must be taken to mitigate the impact of class imbalance (i.e. uneven distribution of approval vs. rejection cases) on classification performance. Special attention is given to improving the model's ability to predict minority classes ("Yes") to support more accurate decision-making and benefit the financial firm.

Key Insights

- Class imbalance i.e. higher rejections than approval significantly affected the model's specificity, and further optimization is needed for predicting the minority class which is the loan approval cases.
- Feature importance analysis revealed that the percentage of income allocated towards loan repayment and annual income of applicant are the most critical variables, highlighting the model's reliance on the relationship between borrowing and income for prediction.

Limitations

- Class Imbalance Issue: Minority class ("Yes") samples are scarce, impacting the model's predictive ability for this class.
- High Feature Dependency: The model heavily relies on a few key features, with lower importance assigned to other features, possibly leading to excessive dependence on these critical features.
- Model Complexity: **XGBoost** has a complex training process and requires considerable time for hyperparameter tuning, which may limit its real-time application capabilities.

Potential Impact:

1. The current model demonstrates excellent classification accuracy and discriminatory power, making it a valuable decision-support tool for improving loan approval and risk management efficiency.
2. Optimizing the model's predictive capability for "high-risk" categories can help reduce false positive rates and enhance business effectiveness.
3. For production deployment, continuous monitoring of model performance and periodic updates are recommended to adapt to changes in data distribution.
4. The big disadvantage arrives from the imbalanced dataset. The model has higher accuracy for rejection cases compared to approval. Thus, a candidate very well eligible for loan approval may be denied the loan due to false negative prediction and thus it is loss for the financial firm as it loses a valuable client.

Future Work

- **Address Class Imbalance:** Use SMOTE or weighted loss functions to enhance the model's predictive performance for the minority class while balancing Sensitivity and Specificity.
- **Feature Enhancement:** Consider introducing more relevant features or adding feature interactions to improve the model's ability to predict anomalous samples.
- **Parameter Optimization:** Current parameters (e.g., $\eta = 0.1$, $\text{max_depth} = 8$, $\text{colsample_bytree} = 0.8$) are optimal within the current parameter space. Further exploration of regularization parameters or broader grid search may yield additional improvements.
- **Optimize Application Scenarios:** With the current accuracy (0.936) and ROC (0.950954), the model is suitable for classification tasks requiring high accuracy (e.g., credit risk assessment). However, it should be used cautiously in scenarios with high specificity requirements.

Final Recommendations

Overall, **XGBoost** performs best, when considering **ROC** and **Accuracy** values, while **random forest** excels in **Sensitivity** and **Kappa**. These predictive models need to be tested on a validation dataset to determine its usability and improve its performance in general. Upon doing that, these models would prove very useful for the financial institutions greatly by reducing the cost, saving time in evaluating and deciding whether to approve or reject a loan application. Unsupervised models are generally trained without the truth labels, however, the DBSCAN clustering done in this project (using the truth labels) are very useful in drawing inferences both from the clusters and the noise points. The noise points can prove particularly useful to understand any anomalies in the future.

Appendix

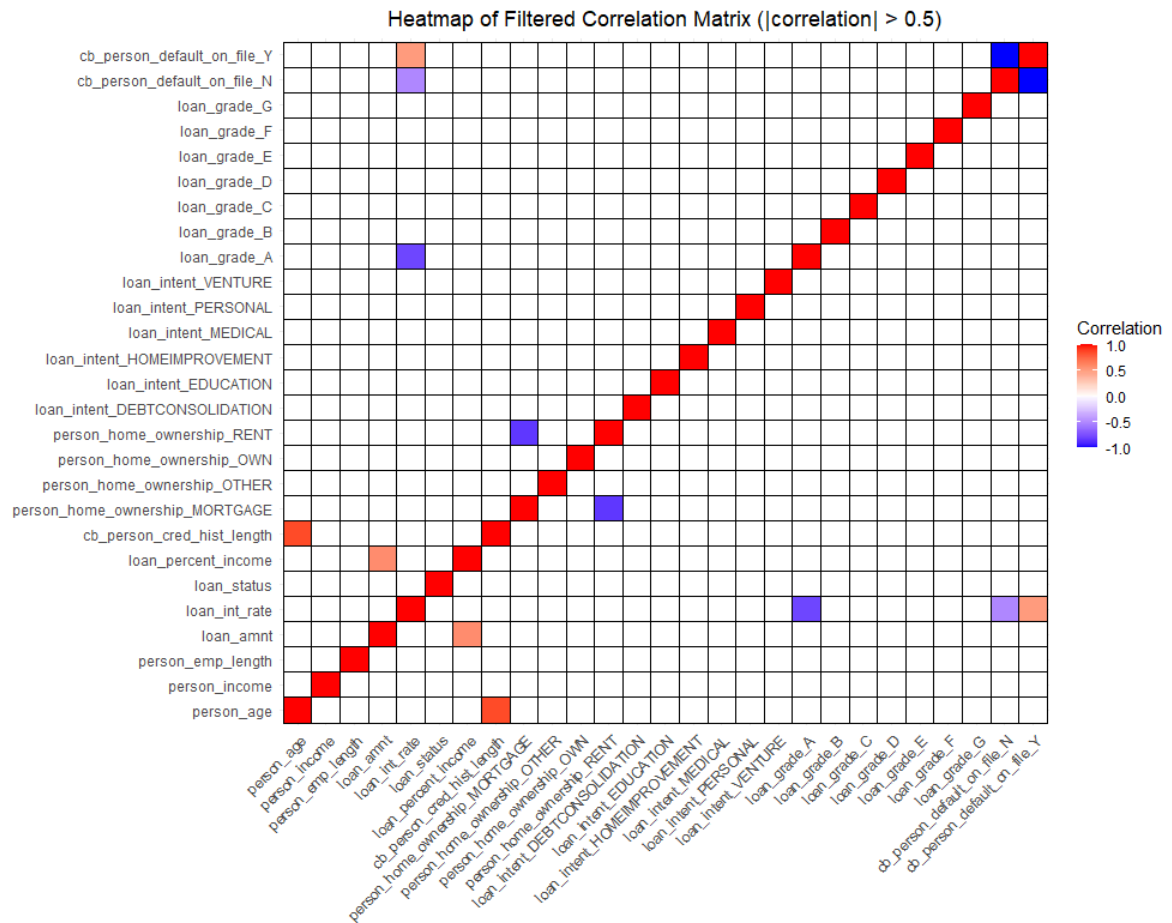
Data Quality Report: Numerical Variables

variable	n	missing	missing_pct	unique	unique_pct	mean	min	Q1	median	Q3	max	sd
person_age	32581	0	0.00	58	0.178	27.735	20.00	23.00	26.00	30.00	144.00	6.348
person_income	32581	0	0.00	4295	13.183	66074.848	4000.00	38500.00	55000.00	79200.00	6000000.00	61983.119
person_emp_length	32581	895	2.75	37	0.114	4.790	0.00	2.00	4.00	7.00	123.00	4.143
loan_amnt	32581	0	0.00	753	2.311	9589.371	500.00	5000.00	8000.00	12200.00	35000.00	6322.087
loan_int_rate	32581	3116	9.56	349	1.071	11.012	5.42	7.90	10.99	13.47	23.22	3.240
loan_status	32581	0	0.00	2	0.006	0.218	0.00	0.00	0.00	0.00	1.00	0.413
loan_percent_income	32581	0	0.00	77	0.236	0.170	0.00	0.09	0.15	0.23	0.83	0.107
cb_person_cred_hist_length	32581	0	0.00	29	0.089	5.804	2.00	3.00	4.00	8.00	30.00	4.055

Data Quality Report: Categorical Variables

variable	n	missing	missing_pct	unique	unique_pct	freqRatio	1st mode	1st mode freq	2nd mode	2nd mode freq	least common	least common freq
person_home_ownership	32581	0	0	4	0.012	1.22	RENT	16446	MORTGAGE	13444	OTHER	107
loan_intent	32581	0	0	6	0.018	1.06	EDUCATION	6453	MEDICAL	6071	HOMEIMPROVEMENT	3605
loan_grade	32581	0	0	7	0.021	1.03	A	10777	B	10451	G	64
cb_person_default_on_file	32581	0	0	2	0.006	4.67	N	26836	Y	5745	Y	5745

Heatmap showing correlations among variables



Supervised Model Parameter Settings

GLM		
1	10-fold cross-validation was configured using trainControl.	method = "cv", number = 10
2	Class probability output was enabled.	classProbs = TRUE
3	A custom evaluation function was implemented.	summaryFunction = custom_summary
4	GLM model based on logistic regression was constructed.	family = binomial(link = "logit")
LASSO		
1	10-fold cross-validation was configured using trainControl.	The same as above
2	Class probability output was enabled.	The same as above
3	A custom evaluation function was implemented.	The same as above
4	A tuneGrid was set to define the search range for the penalty parameter lambda.	The range of lambda was set from 0.001 to 0.1, with a step size of 0.01.
5	The model optimizes based on different lambda values and automatically selects the optimal parameter.	The lambda with the highest ROC value The lambda with the highest accuracy
GBM		
1	10-fold cross-validation was configured using trainControl.	The same as above
2	Each validation randomly splits the data to ensure model performance stability.	
3	Class probability output was enabled.	The same as above
4	A custom evaluation function was implemented.	The same as above
RANGER FOREST		
1	Repeated cross-validation	method = 'repeatedcv', number = 10, repeats = 5
2	Randomly split the data into 10 folds, selecting different training and testing sets for each round.	
3	Class probability output was enabled.	The same as above
4	A custom evaluation function was implemented.	The same as above
5	Perform a random search within the parameter space to find the optimal parameter combination.	search = 'random'
6	Set the number of trees in the forest to 200 to enhance model stability.	num.trees = 200

Phase 1 modelling results

